Prompt Engineering – Assignment

Module: Prompt Engineering | Total marks: 45 (Part A: 15, Part B: 30)

Instructions

- Attempt all questions.
- Use your own words. Do not copy slide text.
- Test every prompt you write in an LLM of your choice. Paste the prompt and the response.
- Part B: choose one option per question. Each question carries 1 mark. No negative marking.

Part A – Prompt Design Exercises (15 marks)

Hands-on Prompt Design

1. Write a vague prompt for summarizing a news article. Rewrite it using all six components. Show both responses and explain the improvement. (5 marks)
2. Design a Few-Shot prompt to classify customer reviews as Positive, Negative or Neutral. Use at least three examples. Test it on two new reviews. (4 marks)
3. Design a Chain-of-Thought prompt for a multi-step word problem. Compare its answer with a plain zero-shot prompt on the same problem. (3 marks)
4. Apply Self-Consistency: run one reasoning prompt five times with temperature above 0. Record the five answers and select the final answer by majority vote. (3 marks)

Part B – Multiple Choice Questions (30 marks)

Q1. Which prompt component defines the persona or expertise the AI should assume?

A. Task B. Role C. Context D. Output Format

Q2. "Summarize the article in no more than 100 words, keeping the key points intact." Which component is this?

A. Role B. Input C. Instruction D. Context

Q3. Which statement about Task and Instruction is correct?

A. Task defines the response layout; Instruction defines the persona. B. Task and Instruction are identical. C. Task states how to do the work; Instruction states what to do. D. Task states what must be accomplished; Instruction specifies how to perform it.

Q4. "Present the recipe in numbered steps with a serving suggestion." Which component is this?

A. Output Format B. Context C. Role D. Input

Q5. In a summarization prompt, the pasted article text "AI is transforming industries by automating tasks…" is the:

A. Role B. Task C. Input D. Context

Q6. "This summary is intended for a general audience with no technical background." Which component is this?

A. Instruction B. Context C. Input D. Task

Q7. Adding "Present the answer in JSON format" to a prompt modifies which component?

A. Instruction B. Task C. Role D. Output Format

Q8. Which statement correctly separates Instruction from Output Format?

A. Instruction describes how to perform the task; Output Format describes how the answer should look. B. Instruction describes how the answer should look; Output Format describes how to perform the task. C. Both mean the same thing. D. Output Format supplies the data the model works on.

Q9. In prompt terminology, a "prompt" is:

A. The tokens the model produces. B. The instructions and context sent to the model. C. The maximum length of the output. D. The total tokens the model can handle at once.

Q10. The context window (max sequence length) is the:

A. Prompt tokens only. B. Output tokens only. C. Total tokens the model handles at once: prompt plus output. D. Number of examples allowed in a prompt.

Q11. What does max_tokens (max_output_tokens) control?

A. The maximum length of the generated output, which must fit inside the context window. B. The size of the context window. C. The number of examples in the prompt. D. The randomness of token selection.

Q12. Which of the following are sampling parameters?

A. Role, Task, Instruction B. Zero-shot, Few-shot, Chain-of-Thought C. Prompt, Input, Context D. Temperature, top-p, max_output_tokens

Q13. A lower temperature produces a probability distribution that is:

A. Flatter, giving more random output. B. Strongly peaked, giving more predictable output. C. Unchanged. D. Larger, giving longer output.

Q14. A higher temperature produces a probability distribution that is:

A. Strongly peaked. B. Shorter in length. C. Flatter, giving more varied output. D. Ignored by the model.

Q15. With Top-N sampling and N = 3, the next token is chosen:

A. Randomly (weighted by probability) from only the 3 most probable tokens. B. Always as the single most probable token. C. From any token in the vocabulary. D. From a list of exactly three prompt words.

Q16. Top-P sampling selects the next token from:

A. A fixed number of top tokens. B. The top tokens whose cumulative probability stays within P. C. All tokens above the temperature value. D. The tokens with the lowest probability.

Q17. Which statement correctly compares Top-N and Top-P?

A. Top-N cuts off by cumulative probability; Top-P cuts off by token count. B. They are identical. C. Top-N cuts off by token count; Top-P cuts off by cumulative probability. D. Top-N controls output length; Top-P controls context window size.

Q18. Prompt Engineering is best described as:

A. Retraining the weights of an LLM. B. The art of designing inputs for LLMs to obtain desired responses. C. Building neural network architectures. D. Compressing models for deployment.

Q19. Zero-shot prompting:

A. Provides several examples to guide the model. B. Breaks the reasoning into intermediate steps. C. Relies on the model's pre-trained knowledge without giving examples. D. Uses an external search tool.

Q20. Few-shot prompting:

A. Gives no examples. B. Runs the same prompt many times. C. Assigns a persona to the model. D. Provides a few examples to guide the desired format and reasoning.

Q21. A prompt shows two labelled reviews ("I love this product!" → Positive; "This is the worst experience ever!" → Negative) and then asks for the label of a new review. Which technique is this?

A. Zero-shot B. Few-shot C. ReAct D. Self-consistency

Q22. Which technique encourages the model to reason through intermediate steps before answering (for example, "Let's think step by step")?

A. Zero-shot prompting B. Role-based prompting C. Chain-of-Thought prompting D. Self-consistency

Q23. A farmer has 3 baskets with 5 apples each and gives away 4 apples. Following Chain-of-Thought reasoning, how many apples remain?

A. 15 B. 11 C. 19 D. 12

Q24. The ReAct cycle follows which sequence?

A. Observation → Reason → Response → Action B. Action → Response → Reason → Observation C. Response → Reason → Action → Observation D. Reason → Action → Observation → Response

Q25. A key limitation of ReAct is that it:

A. Needs external search or tools to obtain live data. B. Cannot perform any reasoning. C. Requires no tools at all. D. Works only on images.

Q26. Which describes Self-Consistency?

A. Running a prompt once with temperature 0. B. Assigning a persona to the model. C. Running the same prompt multiple times with temperature above 0 and choosing the most common answer. D. Giving the model labelled examples.

Q27. In Self-Consistency, who selects the final answer from the multiple outputs?

A. Your own code or process, such as a majority vote outside the model. B. The model decides automatically. C. The context window. D. The top-p setting.

Q28. A main drawback of Self-Consistency is that it:

A. Can be computationally expensive because of multiple runs. B. Needs external search for live data. C. Requires a persona. D. Cannot be used for math problems.

Q29. Which prompt is a better example of Instruction-Based Prompting?

A. "Summarize this article." B. "Tell me about the article." C. "Summarize this article in three bullet points, each under 20 words." D. "Article summary."

Q30. "You are a cybersecurity expert. Explain the importance of multi-factor authentication in banking security." Which technique is this?

A. Few-shot prompting B. Chain-of-Thought prompting C. ReAct D. Role-based prompting